

## DISCUSSION 3

# Joint Distributions and Common Distributions

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$$p_{X,Y}(x, y) \quad f_{Y|X}(y \mid x) \quad X \sim N(\mu, \sigma^2)$$

# Discussion 3 overview

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1. **Calculus review:** basic integrals and integration over a region.
2. **Two random variables:** joint, marginal, and conditional distributions.
3. **Common models:** Bernoulli, binomial, normal, Poisson, and exponential.

## Problem-solving habit

Define the random variable and its support. Translate the event into mathematics. Explain why the formula applies before calculating.

# Calculus review: integrals and double integrals

## Basic integration formulas

$$\int t^n dt = \frac{t^{n+1}}{n+1} + C \quad (n \neq -1),$$

$$\int \frac{1}{t} dt = \ln |t| + C,$$

$$\int e^{ct} dt = \frac{e^{ct}}{c} + C \quad (c \neq 0),$$

$$\int u dv = uv - \int v du.$$

If  $H' = h$ , then

$$\int_a^b h(t) dt = H(b) - H(a).$$

## Double integrals and the region

Integrate in  $y$  first, holding  $x$  constant. For  $0 < x < y$ ,

$$\iint_{0 < x < y} g(x, y) dA = \int_0^\infty \int_x^\infty g(x, y) dy dx.$$

**Example: a fixed  $x > 0$**

$$\begin{aligned} \int_x^\infty 6e^{-x-2y} dy &= 6e^{-x} \left[ -\frac{1}{2}e^{-2y} \right]_x^\infty \\ &= 3e^{-3x}. \end{aligned}$$

**Infinite limits:** integrate to  $M$ , then take  $M \rightarrow \infty$ . Changing integration order requires new bounds for the same region.

# Joint probability mass functions

## Definition

For discrete random variables  $X$  and  $Y$ ,

$$p_{X,Y}(x,y) = P(X = x, Y = y).$$

A valid joint pmf satisfies

$$p_{X,Y}(x,y) \geq 0, \quad \sum_x \sum_y p_{X,Y}(x,y) = 1.$$

For a set  $B$  of possible pairs,

$$P((X,Y) \in B) = \sum_{(x,y) \in B} p_{X,Y}(x,y).$$

## In plain words

Each table entry describes both variables at once. An event can contain several cells.

# Joint pmf: engines and transmissions

## Practice Problem

Let  $X$  and  $Y$  count engines and transmissions requiring repair in one hour. Their joint pmf is

|         | $y = 0$ | $y = 1$ | $y = 2$ | $y = 3$ |
|---------|---------|---------|---------|---------|
| $x = 0$ | .13     | .10     | .07     | .03     |
| $x = 1$ | .12     | .16     | .08     | .04     |
| $x = 2$ | .02     | .06     | .08     | .04     |
| $x = 3$ | .01     | .02     | .02     | .02     |

1. Verify that this is a valid pmf.
2. Find  $P(X = 1, Y = 2)$  and  $P(X + Y = 2)$ .

*Work on the problem before advancing to the solution.*

# Solution: selecting cells in a joint pmf

## Solution

**1. Validity of the pmf.** All entries are nonnegative. The row sums are .33, .40, .20, .07, which add to 1.

**2. The two requested probabilities.** One engine and two transmissions means the single pair (1, 2), so  $P(X = 1, Y = 2) = \boxed{.08}$ .

*Translate a total into pairs.* Exactly two repairs means (0, 2), (1, 1), or (2, 0). These outcomes are mutually exclusive, so the addition rule gives

$$P(X + Y = 2) = .07 + .16 + .02 = \boxed{.25}.$$

**A condition on  $X + Y$  selects a diagonal of the table.**

# Marginal probability mass functions

## Definition

The marginal pmfs describe each variable separately:

$$p_X(x) = \sum_y p_{X,Y}(x, y), \quad p_Y(y) = \sum_x p_{X,Y}(x, y).$$

## In plain words

To keep  $X = x$ , add all ways that  $Y$  can occur. This is the addition rule applied to disjoint events.

## Connection to one-variable probability

After finding  $p_X$ , use it just like any pmf:

$$P(X \in A) = \sum_{x \in A} p_X(x), \quad \mathbb{E}[X] = \sum_x x p_X(x).$$

# Marginal pmfs: repair counts

## Practice Problem

For the same repair-count model,

|         | $y = 0$ | $y = 1$ | $y = 2$ | $y = 3$ |
|---------|---------|---------|---------|---------|
| $x = 0$ | .13     | .10     | .07     | .03     |
| $x = 1$ | .12     | .16     | .08     | .04     |
| $x = 2$ | .02     | .06     | .08     | .04     |
| $x = 3$ | .01     | .02     | .02     | .02     |

find  $p_X(x)$ ,  $p_Y(y)$ , and  $P(Y \geq 2)$ . Explain which direction you sum for each marginal.

*Work on the problem before advancing to the solution.*

# Solution: marginalizing the repair counts

## Solution

Keep  $X$ , sum over  $Y$ . For example,  $p_X(1) = .12 + .16 + .08 + .04 = .40$ . These are the row totals.

| $x$      | 0   | 1   | 2   | 3   |
|----------|-----|-----|-----|-----|
| $p_X(x)$ | .33 | .40 | .20 | .07 |

Keep  $Y$ , sum over  $X$ . The column totals give

| $y$      | 0   | 1   | 2   | 3   |
|----------|-----|-----|-----|-----|
| $p_Y(y)$ | .28 | .34 | .25 | .13 |

Use the marginal pmf.  $P(Y \geq 2) = .25 + .13 = \boxed{.38}$ . Each marginal sums to 1, providing a check.

# Joint probability density functions

## Definition

A joint pdf  $f_{X,Y}$  is nonnegative and integrates to 1:

$$\iint_{\mathcal{S}} f_{X,Y}(x,y) dx dy = 1,$$

where  $\mathcal{S}$  is the support. For an event described by a region  $B$ ,

$$P((X,Y) \in B) = \iint_{B \cap \mathcal{S}} f_{X,Y}(x,y) dx dy.$$

## In plain words

Probability is volume over a region. The support and the event together determine the integration limits. A density height can exceed 1.

# Joint pdf: connection and authentication

## Practice Problem

A server connects at time  $X$  and authenticates at time  $Y$ , both measured in microseconds from the same start. Suppose

$$f_{X,Y}(x,y) = ke^{-x-2y}, \quad 0 < x < y < \infty,$$

and the density is zero elsewhere.

1. Find  $k$  so that this is a valid joint pdf.
2. Find  $P(X < 1, Y < 2)$ .

Describe the allowed values of the inner variable before integrating.

*Work on the problem before advancing to the solution.*

# Solution: normalizing over the support

## Solution to part 1: the normalizing constant

*Read the ordering constraint.* Authentication follows connection. For a fixed  $x > 0$ , the possible authentication times satisfy  $y > x$ .

*Require total probability 1.* Integrate over the full support, including the constraint  $x < y$ :

$$1 = \int_0^\infty \int_x^\infty k e^{-x-2y} dy dx = \int_0^\infty \frac{k}{2} e^{-3x} dx = \frac{k}{6}.$$

*Verify validity.* Since  $e^{-x-2y} > 0$ , nonnegativity requires  $k \geq 0$ . The choice  $k = 0$  cannot integrate to 1, so a valid density needs  $k > 0$ . Our value  $k = 6$  satisfies both nonnegativity and normalization.

**Using  $x > 0, y > 0$  alone would include impossible outcomes with  $y < x$ .**

# Solution: probability over a restricted region

## Solution to part 2: the event probability

*Intersect the event with the support.  $X < 1$  and  $Y < 2$ , together with  $0 < X < Y$ , give  $0 < x < 1$  and  $x < y < 2$ .*

*Integrate over exactly that region.*

$$\begin{aligned}P(X < 1, Y < 2) &= \int_0^1 \int_x^2 6e^{-x-2y} dy dx \\&= \int_0^1 3(e^{-3x} - e^{-x-4}) dx \\&= 1 - e^{-3} - 3e^{-4}(1 - e^{-1}) \approx \boxed{.9155}.\end{aligned}$$

*Check the bounds. With  $y$  outside instead, split at  $y = 1$ : the inner upper limit for  $x$  is  $y$  below 1 and is 1 above 1.*

# Marginal probability density functions

## Definition

Integrate out the variable you do not want:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx.$$

The support determines which part of each integral is nonzero.

## In plain words

Marginalization combines every possible value of the other variable. The resulting formula and support describe only the variable you keep.

# Marginal pdfs: server times

## Practice Problem

For the server model

$$f_{X,Y}(x,y) = 6e^{-x-2y}, \quad 0 < x < y < \infty,$$

find  $f_X(x)$  and  $f_Y(y)$ , including their supports. Then use  $f_Y$  to find  $P(Y < 2)$ .

*Work on the problem before advancing to the solution.*

# Solution: server-time marginals

## Solution

*Keep connection time  $x$ .* Authentication can occur at any  $y > x$ :

$$f_X(x) = \int_x^\infty 6e^{-x-2y} dy = \boxed{3e^{-3x}}, \quad x > 0.$$

*Keep authentication time  $y$ .* Connection must lie between 0 and  $y$ :

$$f_Y(y) = \int_0^y 6e^{-x-2y} dx = \boxed{6e^{-2y}(1 - e^{-y})}, \quad y > 0.$$

Both densities are zero outside their stated supports.

*Return to a one-variable probability.*

$$P(Y < 2) = \int_0^2 (6e^{-2y} - 6e^{-3y}) dy = 1 - 3e^{-4} + 2e^{-6} \approx \boxed{.9500}.$$

# Conditional pmfs and conditional means

## Definition

For  $p_X(x) > 0$ ,

$$p_{Y|X}(y | x) = \frac{p_{X,Y}(x, y)}{p_X(x)}.$$

The conditional mean uses this new pmf:

$$\mathbb{E}[Y | X = x] = \sum_y y p_{Y|X}(y | x).$$

## In plain words

Conditioning on  $X = x$  keeps one row and rescales that row to sum to 1. This is the conditional probability rule from Discussion 2.

# Conditional repair counts

## Practice Problem

In the repair-count model, the row for  $X = 1$  is

| $y$             | 0   | 1   | 2   | 3   |
|-----------------|-----|-----|-----|-----|
| $p_{X,Y}(1, y)$ | .12 | .16 | .08 | .04 |

Given that one engine requires repair, find

1. the conditional pmf of  $Y$ ,
2.  $P(Y > 0 \mid X = 1)$ ,
3.  $\mathbb{E}[Y \mid X = 1]$ .

*Work on the problem before advancing to the solution.*

# Solution: conditioning and renormalizing

## Solution

1. **Conditional pmf of  $Y$ .** First find  $p_X(1) = .12 + .16 + .08 + .04 = .40$ . Divide each entry in the row by .40 so the conditional probabilities sum to 1:

| $y$              | 0   | 1   | 2   | 3   |
|------------------|-----|-----|-----|-----|
| $p_{Y X}(y   1)$ | .30 | .40 | .20 | .10 |

2.  $P(Y > 0 | X = 1)$ . Use the complement under the conditional pmf:

$$P(Y > 0 | X = 1) = 1 - P(Y = 0 | X = 1) = 1 - .30 = \boxed{.70}.$$

3.  $\mathbb{E}[Y | X = 1]$ . Weight each value by its conditional probability:

$$\mathbb{E}[Y | X = 1] = 0(.30) + 1(.40) + 2(.20) + 3(.10) = \boxed{1.10}.$$

# Conditional densities and conditional means

## Definition

For  $f_X(x) > 0$ , the conditional density is

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

Then

$$P(a < Y < b | X = x) = \int_a^b f_{Y|X}(y | x) dy,$$

$$\mathbb{E}[Y | X = x] = \int_{-\infty}^{\infty} y f_{Y|X}(y | x) dy.$$

## In plain words

Fix  $x$  and normalize a slice of the joint density. The denominator is a density value. For continuous  $X$ ,  $P(X = x) = 0$ .

# Conditional density: authentication after connection

## Practice Problem

For the server model,

$$f_{X,Y}(x,y) = 6e^{-x-2y}, \quad 0 < x < y, \quad f_X(x) = 3e^{-3x}, \quad x > 0.$$

1. Find  $f_{Y|X}(y | x)$  and its support.
2. Find  $P(Y > 2 | X = 1.5)$ .
3. Find  $\mathbb{E}[Y | X = 1.5]$ .

All times are in microseconds from the common start.

*Work on the problem before advancing to the solution.*

# Solution: conditional density and a tail probability

## Solution

**1. Conditional density and support.** For each fixed  $x > 0$ , the support requires  $y > x$ :

$$f_{Y|X}(y | x) = \frac{6e^{-x-2y}}{3e^{-3x}} = \boxed{2e^{-2(y-x)}}, \quad y > x.$$

It is zero for  $y \leq x$ , and its integral over  $(x, \infty)$  is 1.

**2. The conditional tail probability.** Given  $X = 1.5$ , authentication must occur after 1.5. The event  $Y > 2$  is the tail above 2:

$$P(Y > 2 | X = 1.5) = \int_2^{\infty} 2e^{-2(y-1.5)} dy = e^{-1} \approx \boxed{.3679}.$$

# Solution: expected authentication time

## Solution to part 3: the conditional mean

*Separate elapsed time from additional waiting.* Set  $u = y - x$ . Then  $u > 0$  and the conditional density becomes  $2e^{-2u}$ .

$$\begin{aligned}\mathbb{E}[Y \mid X = x] &= \int_x^\infty y 2e^{-2(y-x)} dy \\ &= \int_0^\infty (x + u) 2e^{-2u} du \\ &= x + [-ue^{-2u}]_0^\infty + \int_0^\infty e^{-2u} du = x + \frac{1}{2}.\end{aligned}$$

*Interpret the result.*  $\mathbb{E}[Y \mid X = 1.5] = \boxed{2.0 \mu\text{s}}$  from the start. The expected additional wait is  $0.5 \mu\text{s}$ .

**Later, we will recognize the additional wait as exponential with rate 2.**

# The Bernoulli distribution

## Definition

A Bernoulli variable records whether one event occurs:

$$X = \begin{cases} 1, & \text{success,} \\ 0, & \text{failure,} \end{cases} \quad P(X = 1) = p, \quad 0 \leq p \leq 1.$$

Its pmf, mean, and variance are

$$p_X(0) = 1 - p, \quad p_X(1) = p,$$

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p).$$

At  $p = 0$  or  $p = 1$ ,  $X$  is constant and its variance is zero.

## In plain words

A Bernoulli variable is an indicator. The word “success” labels the event we count, even if that event is a defect or failure.

# Bernoulli: a voter indicator

## Practice Problem

In the population described in the notes, 60% of registered voters voted in the last election. Select one person at random and define

$X = 1$  if the person voted,  $X = 0$  otherwise.

Identify the distribution of  $X$ . Derive its mean and variance from the pmf, and interpret the mean.

*Work on the problem before advancing to the solution.*

# Solution: moments of an indicator

## Solution

*Match the variable to the model.*  $X$  has two possible values, 0 and 1, with  $P(X = 1) = .60$ . Therefore  $X \sim \text{Bernoulli}(.60)$ .

*Use the definition of expectation.*

$$\mathbb{E}[X] = 0(.40) + 1(.60) = \boxed{.60}.$$

*Use the identity  $X^2 = X$ .*

$$\mathbb{E}[X^2] = .60, \quad \text{Var}(X) = .60 - .60^2 = \boxed{.24}.$$

*Interpret.* The mean of a 0–1 indicator equals the probability of the event. This is why an average of indicators is a sample proportion.

# The binomial distribution

## Definition

If  $I_1, \dots, I_n$  are independent Bernoulli( $p$ ) variables, then

$$X = \sum_{i=1}^n I_i \sim \text{Binomial}(n, p).$$

The pmf is

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, \dots, n.$$

$$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1-p).$$

## In plain words

Check a fixed number of trials, two outcomes per trial, a common success probability, and independence. The coefficient counts which trials succeed.

# Binomial: on-time flights

## Practice Problem

In the model used in the notes, a flight arrives on time with probability .89. Assume the on-time indicators for 12 selected flights are independent. Let  $X$  count on-time arrivals.

1. Explain why  $X$  is binomial and find  $P(X = 10)$ .
2. Find  $P(X \geq 10)$  and  $P(X \leq 11 \mid X \geq 10)$ .
3. Find  $\mathbb{E}[X]$  and  $\text{SD}(X)$ .

Would common weather delays make the independence assumption questionable?

*Work on the problem before advancing to the solution.*

# Solution: binomial probabilities and conditioning

## Solution

**1. Model and exactly 10 arrivals.** The 12 trials are independent with binary outcomes and common success probability .89. Thus  $X \sim \text{Binomial}(12, .89)$ .

*Counting.* There are  $\binom{12}{10}$  arrangements, each with probability  $.89^{10}.11^2$ :

$$P(X = 10) = \binom{12}{10} .89^{10} .11^2 \approx \boxed{.2490}.$$

**2. Tail and conditional probabilities.**

$$P(X \geq 10) = \sum_{k=10}^{12} \binom{12}{k} .89^k .11^{12-k} \approx \boxed{.8623},$$

$$P(X \leq 11 \mid X \geq 10) = \frac{P(X = 10) + P(X = 11)}{P(X \geq 10)} \approx \boxed{.7136}.$$

# Solution: binomial moments and independence

## Solution

**3. Mean and standard deviation.** Let  $I_i$  indicate an on-time arrival. Since  $X = \sum_i I_i$ , linearity gives

$$\mathbb{E}[X] = \sum_i \mathbb{E}[I_i] = 12(.89) = \boxed{10.68}.$$

*Use independence for the variance.* The covariance terms vanish, so

$$\text{Var}(X) = \sum_i \text{Var}(I_i) = 12(.89)(.11) = 1.1748, \quad \text{SD}(X) \approx \boxed{1.084}.$$

The standard deviation is measured in flights.

*Weather question.* Common weather can make arrivals dependent. Then the binomial pmf and variance need not apply. Linearity of expectation still applies.

# The normal distribution and standardization

## Definition

If  $X \sim N(\mu, \sigma^2)$  with  $\sigma > 0$ , then

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \quad -\infty < x < \infty.$$

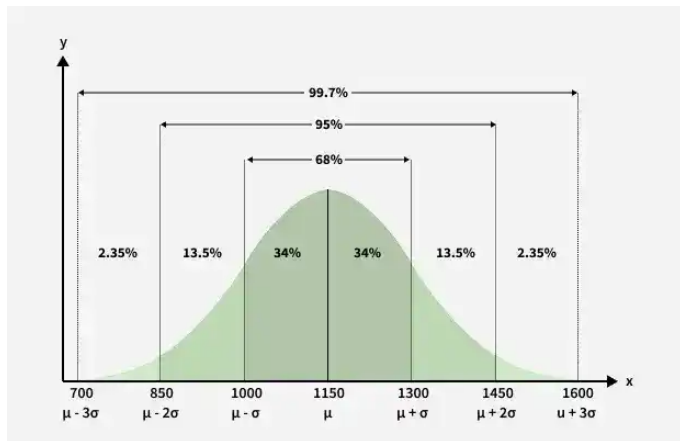
The mean is  $\mu$  and the variance is  $\sigma^2$ . Standardization gives

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1), \quad \Phi(z) = P(Z \leq z).$$

## In plain words

Subtract the center, then divide by the standard deviation. A  $z$  value measures distance from the mean in standard-deviation units. Standardizing an arbitrary variable does not make it normal.

# The normal bell curve



This illustration uses  $\mu = 1150$  and  $\sigma = 150$ .

**Standardization preserves probabilities for corresponding intervals.**

## Reading the curve

### Center and spread

$$z = \frac{x - \mu}{\sigma}.$$

At  $x = \mu$ ,  $z = 0$ . At  $x = \mu \pm \sigma$ ,  $z = \pm 1$ .

### Area is probability

$$P(|Z| \leq 1) \approx 68\%,$$

$$P(|Z| \leq 2) \approx 95\%,$$

$$P(|Z| \leq 3) \approx 99.7\%.$$

# Normal probabilities: reaction times

## Practice Problem

Suppose reaction time  $R$  to brake lights follows a normal model with mean 1.25 seconds and standard deviation .46 second.

1. Find  $P(1.00 < R < 1.75)$ .
2. This model assigns some probability to negative reaction times. Find  $P(R < 0)$  and comment on this limitation.

Use  $\Phi(z) = P(Z \leq z)$  from a standard normal table or calculator.

*Work on the problem before advancing to the solution.*

# Solution: standardizing both endpoints

## Solution

1. **Interval probability.**  $R \sim N(1.25, .46^2)$ . Divide by  $\sigma = .46$ .

$$\begin{aligned} P(1 < R < 1.75) &= P\left(\frac{1 - 1.25}{.46} < Z < \frac{1.75 - 1.25}{.46}\right) \\ &= \Phi(1.08696) - \Phi(-.54348) \approx \boxed{.5681}. \end{aligned}$$

A table using  $z = 1.09$  and  $z = -.54$  gives  $.8621 - .2946 = .5675$ . The small difference is due to rounding.

2. **Negative reaction times and model limitation.**

$$P(R < 0) = \Phi\left(\frac{-1.25}{.46}\right) \approx \boxed{.0033}.$$

Negative times are impossible. The normal approximation places a small probability outside the physical support.

# The Poisson distribution

## Definition

A Poisson variable counts events in a fixed interval:

$$N_t \sim \text{Poisson}(\lambda t), \quad P(N_t = k) = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \quad k = 0, 1, \dots$$

$$\mathbb{E}[N_t] = \text{Var}(N_t) = \lambda t, \quad \text{SD}(N_t) = \sqrt{\lambda t}.$$

Here  $\lambda$  is a rate per unit of time or space, and  $t$  is the interval length.

## In plain words

For a homogeneous Poisson process, disjoint intervals have independent counts, the rate is constant, and multiple events in a sufficiently short interval have negligible probability compared with its length.

# Poisson: potholes along a highway

## Practice Problem

Assume potholes follow a homogeneous Poisson process along a highway, averaging 7 per 10 miles.

1. Find the probability of exactly 8 potholes in 10 miles.
2. Find the probability of exactly 8 potholes in 20 miles.
3. Find the mean and standard deviation of the count in 70 miles.

Identify the rate and the expected count for each interval before applying a formula.

*Work on the problem before advancing to the solution.*

# Solution: rate versus expected count

## Solution

*Rate and units.*  $\lambda = .7$  potholes per mile, so the mean count in  $t$  miles is  $.7t$ .

**1. Exactly 8 potholes in 10 miles.** Use  $\lambda t = 7$ :

$$P(N_{10} = 8) = e^{-7} \frac{7^8}{8!} \approx \boxed{.1304}.$$

**2. Exactly 8 potholes in 20 miles.** Use  $\lambda t = 14$ :

$$P(N_{20} = 8) = e^{-14} \frac{14^8}{8!} \approx \boxed{.0304}.$$

**3. Mean and standard deviation in 70 miles.** Since  $N_{70} \sim \text{Poisson}(49)$ , the mean and variance both equal 49:

$$\mathbb{E}[N_{70}] = \boxed{49}, \quad \text{SD}(N_{70}) = \sqrt{49} = \boxed{7}.$$

# The exponential distribution

## Definition

Let  $T$  be the time to the first event in a homogeneous Poisson process with rate  $\lambda > 0$ . Then  $T \sim \text{Exponential}(\lambda)$ , with

$$f_T(t) = \lambda e^{-\lambda t}, \quad t \geq 0, \quad F_T(t) = 1 - e^{-\lambda t}, \quad t \geq 0.$$

Both are zero for  $t < 0$ . Its survival probability and moments are

$$P(T > t) = e^{-\lambda t}, \quad \mathbb{E}[T] = \frac{1}{\lambda}, \quad \text{Var}(T) = \frac{1}{\lambda^2}.$$

## In plain words

The parameter is a rate. A higher rate means a shorter average wait. The same model can describe distance between events.

# Exponential and Poisson: flaws in a cable

## Practice Problem

Assume flaws follow a homogeneous Poisson process along a cable. The mean distance between consecutive flaws is 12 m. Let  $T$  be the distance from a fixed starting point to the next flaw and  $N_t$  the number of flaws in  $t$  meters.

1. Find the rate  $\lambda$  and  $P(T > 15)$ .
2. Compute  $P(N_{15} = 0)$  and explain its connection to part 1.
3. Find  $P(8 < T < 20)$ .

*Work on the problem before advancing to the solution.*

# Solution: one process, two random variables

## Solution

**1. Rate and waiting beyond 15 m.** The mean distance is  $1/\lambda = 12$  m, so  $\lambda = 1/12$  flaws per meter. The exponential survival function gives

$$P(T > 15) = e^{-15/12} \approx \boxed{.2865}.$$

**2. No flaws in 15 m.** Use the Poisson pmf with mean  $15/12$ :

$$P(N_{15} = 0) = e^{-15/12} \frac{(15/12)^0}{0!} \approx \boxed{.2865}.$$

Waiting beyond 15 m means encountering no flaw in those 15 m, so  $\{T > 15\} = \{N_{15} = 0\}$ .

**3. Distance between 8 and 20 m.** Subtract cumulative probabilities:

$$P(8 < T < 20) = F_T(20) - F_T(8) = e^{-8/12} - e^{-20/12} \approx \boxed{.3245}.$$

# The memoryless property

## Definition

If  $T \sim \text{Exponential}(\lambda)$ , then for  $s, t \geq 0$ ,

$$P(T > s + t \mid T > s) = P(T > t) = e^{-\lambda t}.$$

## In plain words

Once we know an exponential waiting time has exceeded  $s$ , the additional waiting time has the original exponential distribution.

## Connection to conditional probability

Since  $\{T > s + t\} \subseteq \{T > s\}$ ,

$$P(T > s + t \mid T > s) = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t}.$$

# Memorylessness: pore diameters

## Practice Problem

As in the notes, suppose a pore diameter  $D$  in microns follows  $\text{Exponential}(.25)$ .

1. Find  $P(D > 11 \mid D > 3)$ .
2. Find the median diameter.

For part 1, identify the additional distance beyond 3. For part 2, state the cumulative probability that defines a median.

*Work on the problem before advancing to the solution.*

# Solution: conditional tails and a median

## Solution

1. **The conditional tail probability.** The additional amount is  $11 - 3 = 8$  microns:

$$P(D > 11 \mid D > 3) = \frac{e^{-.25(11)}}{e^{-.25(3)}} = e^{-.25(8)} \approx \boxed{.1353}.$$

This equals  $P(D > 8)$  by memorylessness.

2. **Median diameter.** The median  $m$  satisfies  $P(D \leq m) = .5$ :

$$1 - e^{-.25m} = .5 \quad \implies \quad m = \frac{\ln 2}{.25} \approx \boxed{2.773 \text{ microns}}.$$

*Compare center measures.* The mean is  $1/.25 = 4$  microns. The longer right tail makes the mean exceed the median.

# Choosing a probability model

| What does the variable record?                                            | Candidate model           |
|---------------------------------------------------------------------------|---------------------------|
| One binary outcome                                                        | Bernoulli( $p$ )          |
| Successes in $n$ independent trials with common $p$                       | Binomial( $n, p$ )        |
| A measurement modeled by a symmetric bell-shaped density                  | Normal( $\mu, \sigma^2$ ) |
| Events in an interval of length $t$ under a constant-rate Poisson process | Poisson( $\lambda t$ )    |
| Waiting time to the next event in that process                            | Exponential( $\lambda$ )  |

**The meaning of the random variable and the assumptions determine the model.**

# Common distribution reference

| Model                    | Support           | Mean        | Variance      |
|--------------------------|-------------------|-------------|---------------|
| Bernoulli( $p$ )         | $\{0, 1\}$        | $p$         | $p(1 - p)$    |
| Binomial( $n, p$ )       | $\{0, \dots, n\}$ | $np$        | $np(1 - p)$   |
| $N(\mu, \sigma^2)$       | $\mathbb{R}$      | $\mu$       | $\sigma^2$    |
| Poisson( $\lambda t$ )   | $\{0, 1, \dots\}$ | $\lambda t$ | $\lambda t$   |
| Exponential( $\lambda$ ) | $[0, \infty)$     | $1/\lambda$ | $1/\lambda^2$ |

## Checks before calculating

Does the variable count events or measure a quantity? Do the assumptions fit? Are the rate and interval units consistent? Does the event require a sum, an integral, a complement, or a conditional probability?